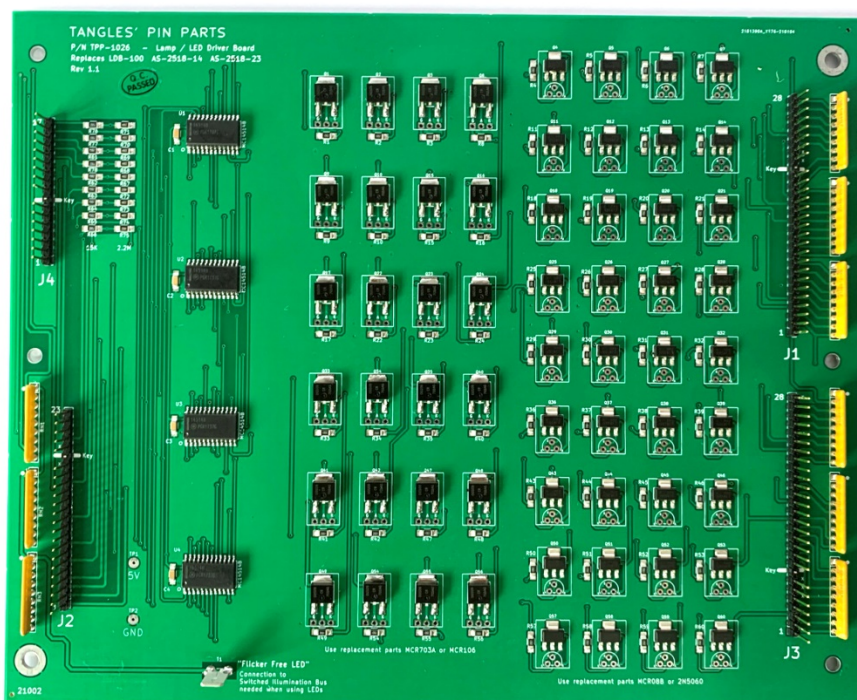


TANGLES' PIN PARTS

TPP-1026

Lamp / LED Driver Board For Bally / Stern Systems

Replacement for AS-2518-14, AS-2518-23 and LDB-100



Description

The Tangles Lamp / LED Driver (TPP-1026) is a re-designed driver board for use in Bally and Stern solid state pinball machines.

The bonus feature of this board is its ability to drive LEDs and standard incandescent lamps!

As always, Tangles' boards use top tier parts sourced through reliable supply chains. This board utilises modern components and manufacturing to extend the life and reliability of your Pinball machine.

Features

- Supports flicker free LED operation.
- Supports standard (incandescent) lamps.
- High quality gold contact on long pins make for a reliable connection to the wiring harness.
- A direct replacement for the original boards. Uses the same mounting holes and connections.
- Supports surface mount and through hole replacement thyristors.
- No soldering required.

Installation

Warning: High voltages are present inside the pinball machine. Turn off the power and unplug the mains before you commence work.

1. Ensure power is off.
2. Open the back cabinet and unplug the 4 connectors.
3. Remove the 2 securing screws (Bottom left and top right)
4. Unclip the 4 mounting posts and remove the old board.
5. Install the TPP-1026 onto the 4 mounting posts (push on)
6. Secure the two mounting screws.
7. Reconnect the 4 Molex connectors taking care to align the "Key" with the missing pin on plug.

Optional: For LED Operation

8. Connect the supplied jump cable to the FASTON connection T1 on the board.
9. Attach the other end of this cable to the "Switched Illumination Bus" (See section below)

Attaching to Switched Illumination Bus

For the TPP-1026 to provide flicker free LED operation, a connection is needed from the FASTON plug T1 to the game's switched illumination bus.

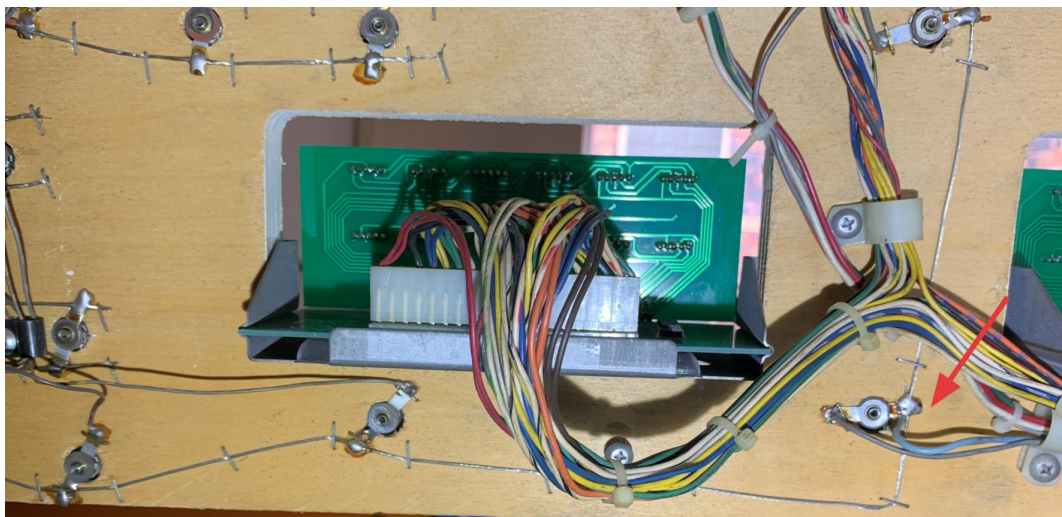
Your board ships with a pre-made cable that has a FASTON connector on one end and an alligator clip on the other.

The best place to attach to the Switched Illumination bus (5.4v DC) is on the door of the back box. Most of the lights on the back-box door are General-Illumination (GI) but some will be SWITCHED LAMPS. Like the Player 1 lamp, Tilt lamp, High Score lamp etc. These lamps are wired with a thin striped colour wire on the control side, and a solid colour thicker wire on the other. This thick wire is the Switched Illumination Bus.

Computer controlled lamps turn off and on under the control of the CPU. They have individual striped wires on one side and are often joined together on the "Switched Illumination Bus" side with a bare wire.

Connect the alligator clip to this wire.

Route the jump wire neatly with the other head board wires then connect the FASTON connector onto terminal T1 of the TPP-1026.



In this photo, the three lower lamps and the upper right lamp are computer controlled lamps. The bare wire is the Switched Illumination bus. The thicker, solid colour wire (blue) is seen on the bottom right of the picture.

Note: Attaching this wire is only necessary if you are using LEDs

Game Compatibility

The **Tangles LAMP / LED Driver Board** is compatible with the following games:

STERN

Ali, Big Game, Catacomb, Cheetah, Cosmic Princess, Dracula, Dragonfist, Flight 2000, Freefall, Galaxy, Iron Maiden, Lazer Lord, Lectrnamo, Lightning, Magic, Memory Lane, Meteor, Nine Ball, Nugent, Orbitor 1, Pinball, Quicksilver, Seawitch, Split Second, Star Gazer, Stars, Stingray, Trident, Viper, Wild Fyre

BALLY

Black Jack, Black Pyramid, BMX, Centaur, Centaur II, Cybernaut, Dolly Parton, Eight Ball, Eight Ball Deluxe, Eight Ball Deluxe Ltd, Elektra, Embryon Mystic, Evil Knievel, Fathom, Fireball Classic, Fireball II, Flash Gordon, Freedom, Frontier, Future Spa, Grand Slam, Harlem Globe Trotters, Hotdoggin, Kings of Steel, Kiss, Lost World, Mata Hari, Medusa, Mr. and Mrs. Pacman, Night Rider Vector, Nitro Ground Shaker, Paragon, Playboy, Power Play, Rapid Fire, Rolling Stone, Silver Ball Mania, Six Million Dollar Man, Skate Ball, Space Invaders, Speakeasy, Spectrum, Spy Hunter, Star Trek, Strikes and Spares, Super Sonic, Viking, Voltan, Xenon, X's and O's

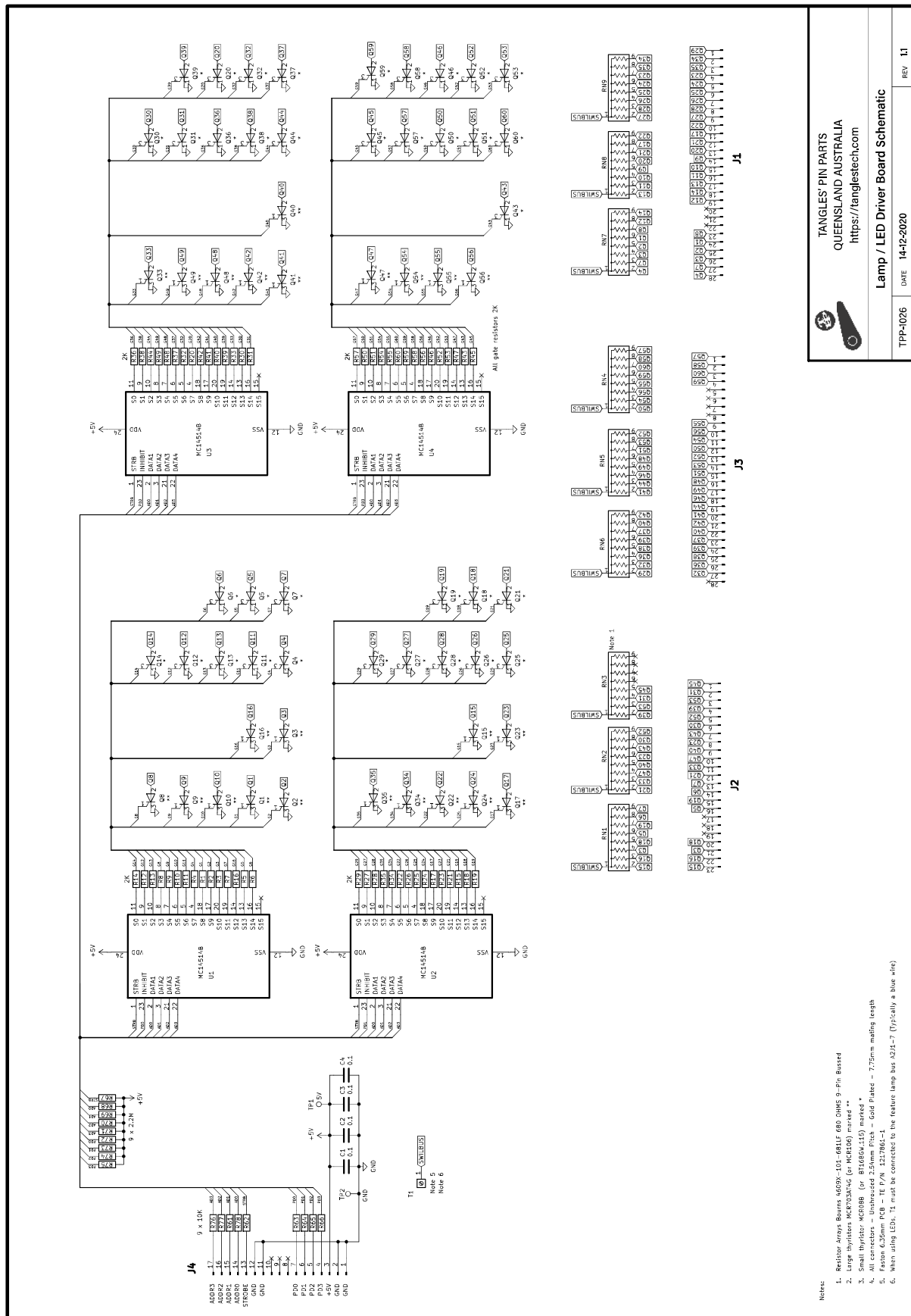
Trouble Shooting

Problem	Comments
No switched lights are working.	• Check Fuse F1 (typically 10A) on the Rectifier board. • Using a multimeter, check for the presence of 5VDC between TP1 (5v) and TP2 (Ground) • Check the two screw mounts are secured to provide ground to the board.
LEDs lamps are flickering	• Check the wire is installed correctly connecting the TPP-1026 T1 connection to the switched illumination bus.

Testing

Follow the instruction from your game manual for LAMP testing. This is typically done by:

- TURN ON POWER
- PRESS SELF TEST BUTTON ON INSIDE OF FRONT DOOR TO ENTER LAMP DRIVER MODULE TEST
- ALL FEATURE LAMPS FLASH ON AND OFF.
- TURN POWER OFF THEN ON. GAME READY TO PLAY



TANGLES' PIN PARTS
QUEENSLAND AUSTRALIA
<https://tanglestech.com>

Lamp / LED Driver Board Schematic

TPP-1026 DATE 14-12-2020 REV 11

Support and Other Resources

In the event that you require support for your product or assistance with installation please contact Tangles directly at tangles@tanglestech.com

Video illustrating the installation:

<http://bit.ly/TPP1026VID>

Updates to this manual will be posted here:

<http://bit.ly/TPP1026>